

Davide Ermellino

Cagliari, Italy | +39 349 410 3867 | ermellinodavide@gmail.com | linkedin.com/in/davide-ermellino | github.com/Davidermellino
Italian citizen. Authorized to work in the EU and EEA without sponsorship.

MSc Data Science Student, Machine Learning and Backend Engineering

Data Science master's student graduating July 2028 and returning to study after summer 2027. Three research and industry roles writing production Python: LLM code-generation agents, computer vision pipelines and geospatial data infrastructure. Seeking a summer 2027 software engineering internship in the EU.

Education

EIT Digital Master School, Madrid ES and Budapest HU

09/2026 - 07/2028

M.S. Data Science, double degree. Year 1 Universidad Politecnica de Madrid, Year 2 Eotvos Lorand University. Minor in Innovation and Entrepreneurship.

- Enrolled full time, returning to study after summer 2027. Expected graduation July 2028.
- Coursework: Deep Learning, Information Retrieval, Cloud Computing and Big Data Ecosystems Design, Network Science, Stream Mining.

University of Cagliari, Cagliari IT

07/2026

B.S. Applied Computer Science and Data Analytics. Final grade 110/110 with honours (cum laude), the top mark in the Italian system.

- Coursework: Data Structures and Algorithms, Operating Systems, Software Engineering, Machine Learning, Artificial Intelligence, Big Data.

Skills

Languages: Python | C | SQL | JavaScript | R

Frameworks and libraries: TensorFlow | PyTorch | OpenCV | Ultralytics YOLO | EasyOCR | scikit-learn | LangChain | React | D3.js

Tools and infrastructure: PostgreSQL | PostGIS | Git | Unity | Tkinter

Work Experience

RE/MAX | Data Scientist

04/2026 - 07/2026

- Built the Python and PostGIS ETL pipeline that feeds the company's property valuation models, normalizing demographic, geospatial and pricing records for all 308 Portuguese municipalities into one queryable dataset.
- Wrote the spatial joins and geometry validation in EPSG:3763 that reconcile separate national data sources onto a shared municipality key.
- Shipped a JavaScript and D3.js front end over the pipeline output so analysts could inspect regional trends and model features directly.

University of Cagliari | Undergraduate Researcher

01/2026 - 06/2026

- Raised compilation rate 53% over the unvalidated baseline by building a Python validation layer for an LLM multi-agent pipeline that generates Unity C# code.
- Wrote the automated evaluation harness: a ground-truth dataset over planner and executor outputs, gated on a Roslyn compile check, then scored by CodeBLEU and template precision and recall.
- Benchmarked three validator models across zero-shot, few-shot and skills-based prompting to find the configuration that converged in the fewest feedback rounds.
- First author on Learner and Instructor Perceptions of Teaching Industrial Robotics with Artificial Intelligence and Extended Reality, under review at SIREM 2026.

Universidad de Las Palmas de Gran Canaria | Undergraduate Researcher

08/2025 - 01/2026

- Reached 98% digit accuracy on runner bib recognition by pairing YOLO detection with a vision-language model for the read step, across multiple camera views.
- Replaced manual bib logging for race organizers and cut identification time 80%.

Projects

- **Real-time red-light violation detection** [github.com/Davidermellino/real-time-red-light-violation-detection] - Multithreaded Python pipeline that flags vehicles crossing the stop line on red. Four threads pass work through shared buffers: frame capture, batched Ultralytics YOLO detection of light state, stop line and plates, EasyOCR text extraction, and a non-blocking main loop that overlays results and logs violations to CSV.
- **Reinforcement learning on PandaGym** [github.com/Davidermellino/rl-with-panda-gym] - Implemented DQN with and without a target network, REINFORCE in discrete and continuous form, and DDPG with Hindsight Experience Replay in TensorFlow. DDPG with HER solved both PandaReach and PandaPush. Packaged and published on PyPI as rl-with-panda-gym.
- **DiabetesPredictor** [github.com/Davidermellino/DiabetesPredictor] - Tkinter desktop app for diabetes risk prediction. Wrote KNN and random forest from scratch and benchmarked them against scikit-learn decision tree, naive Bayes and neural network, behind a preprocessing layer for feature selection, scaling and SMOTE versus undersampling.

Awards and Accolades

- 2026 | LeadTheFuture Mentee, selected for a STEM mentorship non-profit that admits under 15% of applicants.